

Problem Set 9 — Systems & Inclined Planes

Physics 2026-27 · due at the start of class, Monday, November 16 · Unit 2 Test
Thu, Nov 19

NAME _____

DATE _____

How to work this set. Systems: use the *system view* (total mass, external forces only) to find a , then one object's *individual FBD* to find T . Inclines: tilt your axes with the ramp — parallel component $mg \sin\theta$, perpendicular $mg \cos\theta$, and $N = mg \cos\theta$, not mg . When in doubt, sanity-check your trig at $\theta = 0^\circ$ and 90° .

1 Two blocks, one string

SYSTEMS

A 6.0 kg block and a 4.0 kg block sit on a frictionless table, connected by a string; a 30 N force pulls the 6.0 kg block (and the system) forward.

- System view: find the acceleration.
- Individual view: isolate the trailing 4.0 kg block and find the tension.
- Check yourself on the 6.0 kg block: does $F - T = m_1 a$ hold with your numbers?

K	EVERY GIVEN, WITH UNITS
D	A FIRST, THEN T
E	SYSTEM: $F = (M_1 + M_2)A$ · INDIVIDUAL: $T = M_2 A$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

2 The Atwood machine

SYSTEMS

A 7.0 kg mass and a 5.0 kg mass hang from opposite ends of a string over a frictionless, massless pulley.

- Which way does the system move? Say so before any algebra.
- Find the acceleration. (The driving force is the *difference* in the weights.)
- Find the tension. Then check: is T between the two weights? It has to be — say why.

K	EVERY GIVEN, WITH UNITS
D	A, THEN T
E	$(M_1 - M_2)G = (M_1 + M_2)A$ · THEN ONE MASS ALONE
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

3 Frictionless ramp

INCLINES

A 3.0 kg block is released on a frictionless 25° incline.

- Find its acceleration down the slope. (Notice what cancels.)
- Find the normal force on the block.
- A 30 kg block is released beside it. Which reaches the bottom first? Use your answer to (a).

K	EVERY GIVEN, WITH UNITS
D	THE TARGETS & THEIR UNITS
E	$A = G \sin \theta$ · $N = MG \cos \theta$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

4 Ramp with friction

INCLINES

A block slides down a 35° incline with $\mu_k = 0.30$. Find its acceleration. Mass isn't given — show why you don't need it.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	$A = G(\sin \theta - \mu_k \cos \theta)$
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

5 Does it stay put?

STATIC CASE

A crate sits on a 15° loading ramp with $\mu_s = 0.30$. Does it slide on its own, or stay put? Use the $\tan\theta$ vs. μ_s comparison from class — and show in one line of algebra where that comparison comes from.

WORK

6 The capstone combo

CHALLENGE

A 6.0 kg block on a frictionless 30° incline is connected by a string over a pulley to a 4.0 kg mass hanging off the top edge.

- Tug-of-war first: compare m_2g to $m_1g \sin\theta$. Which side wins, and which way does the system move?
- Find the acceleration (system view — T is internal).
- Find the tension (isolate the hanging mass).

K	EVERY GIVEN, WITH UNITS
D	A, THEN T
E	$M_2G - M_1G \sin\theta = (M_1 + M_2)A$
A	SYMBOLS FIRST • UNITS THE WHOLE WAY

7 Sine or cosine?

SANITY CHECK

A classmate writes $N = mg \sin\theta$ for a block on a ramp. Without touching a calculator, use the $\theta = 0^\circ$ check (flat ground) to prove them wrong in two sentences.

ANSWER

Graded on the homework 4–3–2–1 scale. This is the last problem set before the **Unit 2 Test (Thu, Nov 19)** — the combo in problem 6 is exactly the shape of the test's hardest item.